

Tempest: Porewater Nutrients

2025 Samples

2026-05-26

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0.1 Run Information

```
cat("Run Information: NAME ") #lets you know what section you're in
```

```
## Run Information: NAME
```

```
#set the run date & user name
```

```
sample_year <- "2025"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accomodate ultiple runs in a month
```

```
NH3_PO4_files<- c("Raw Data/20260504_TMP2025_NP_Run3.csv", "Raw Data/20260508_TMPmix_NP_Run1.csv")
```

```
NOx_files <- c("Raw Data/20260504_TMP2025_NOx_Run3.csv", "Raw Data/20260508_TMPmix_NOx_Run1.csv")
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
file_path <- "Raw Data/SEAL_TEMPEST_QAQC_Log_2024.csv"
```

```
final_path <- "Processed Data/Tempest_Nutrients_2025_Run3_Runmix.csv"
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- ""
```

```
#Set up file path for metadata
```

```
#downloaded metadata csv - downloaded from Google drive as csv for this year
```

```
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2024.csv"
```

```
cat(run_notes)
```

```
##Setup
```

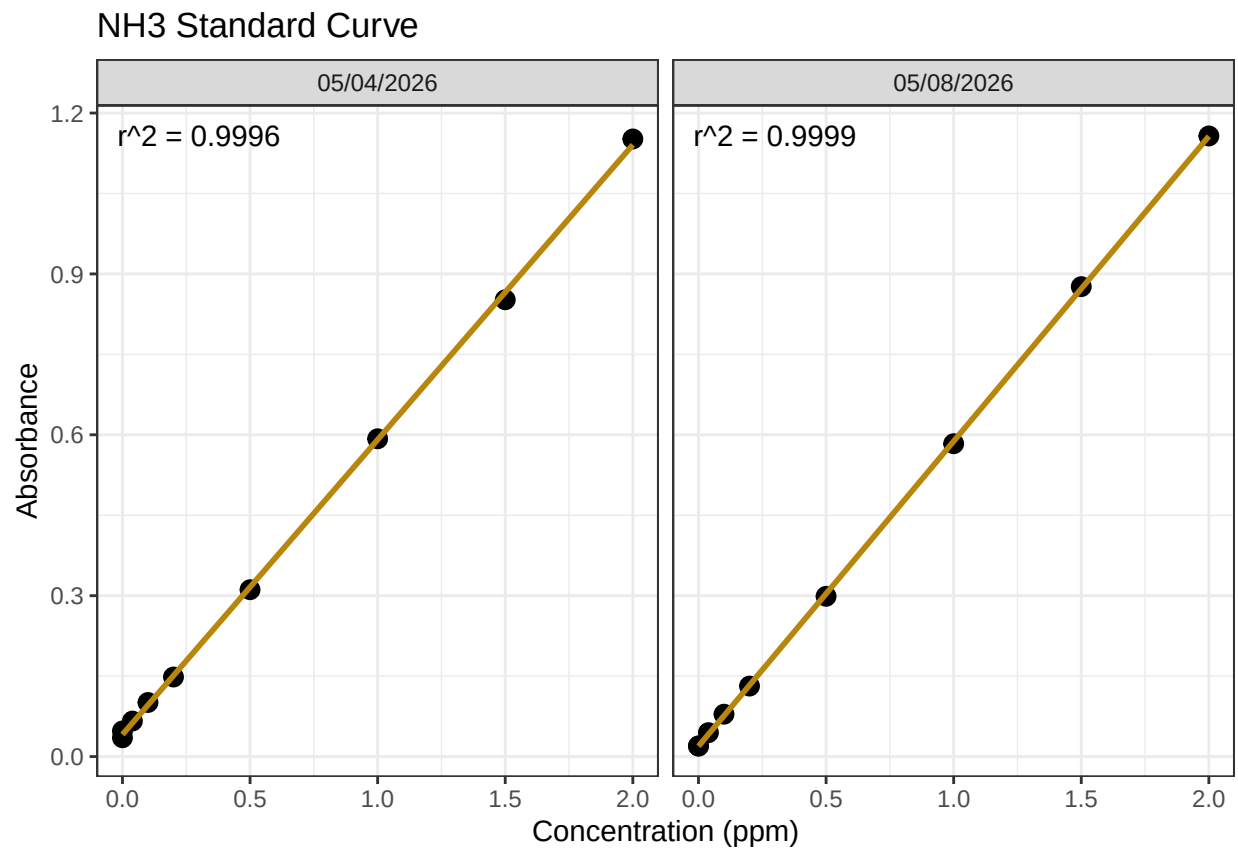
```
##Pull in active porewater tracking inventory sheet from Google Drive:
```

```
##Create similar sample IDs to match with run samples
```

```
##Import Data & Clean
```

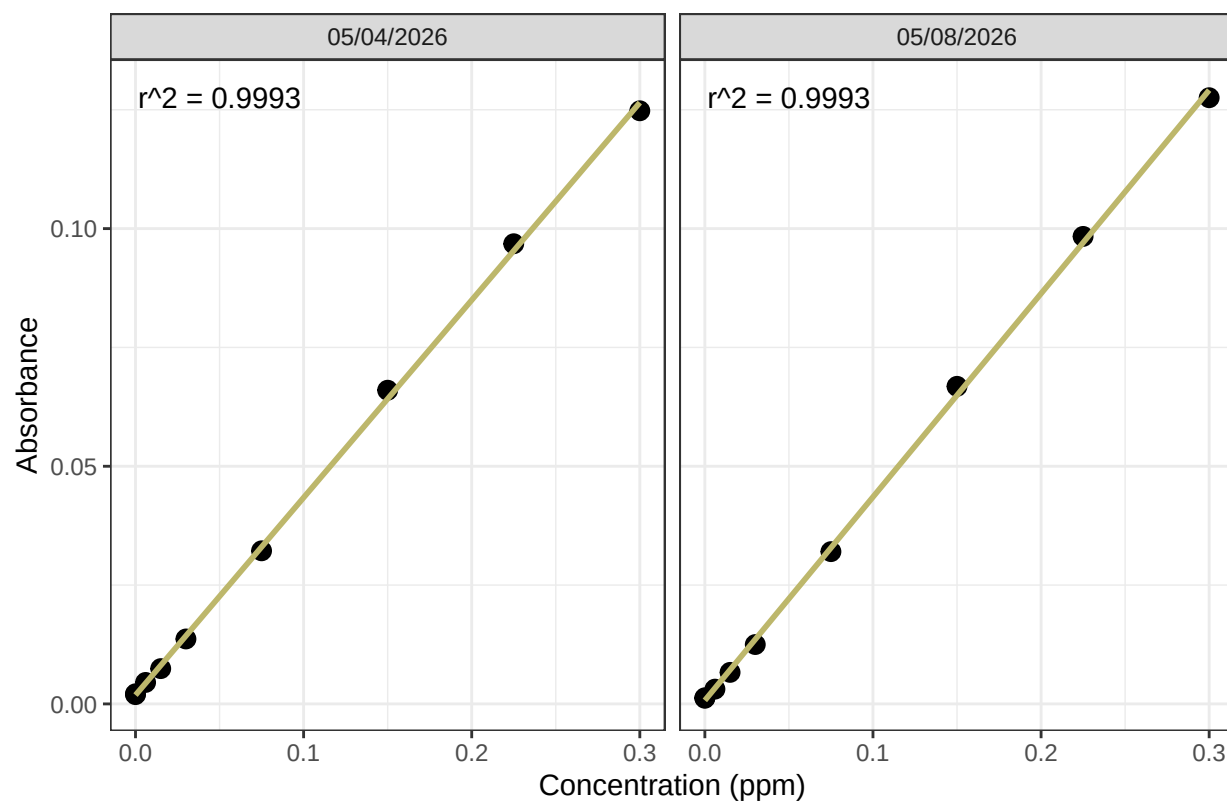
0.2 Assessing standard Curves

```
## `geom_smooth()` using formula = 'y ~ x'
```

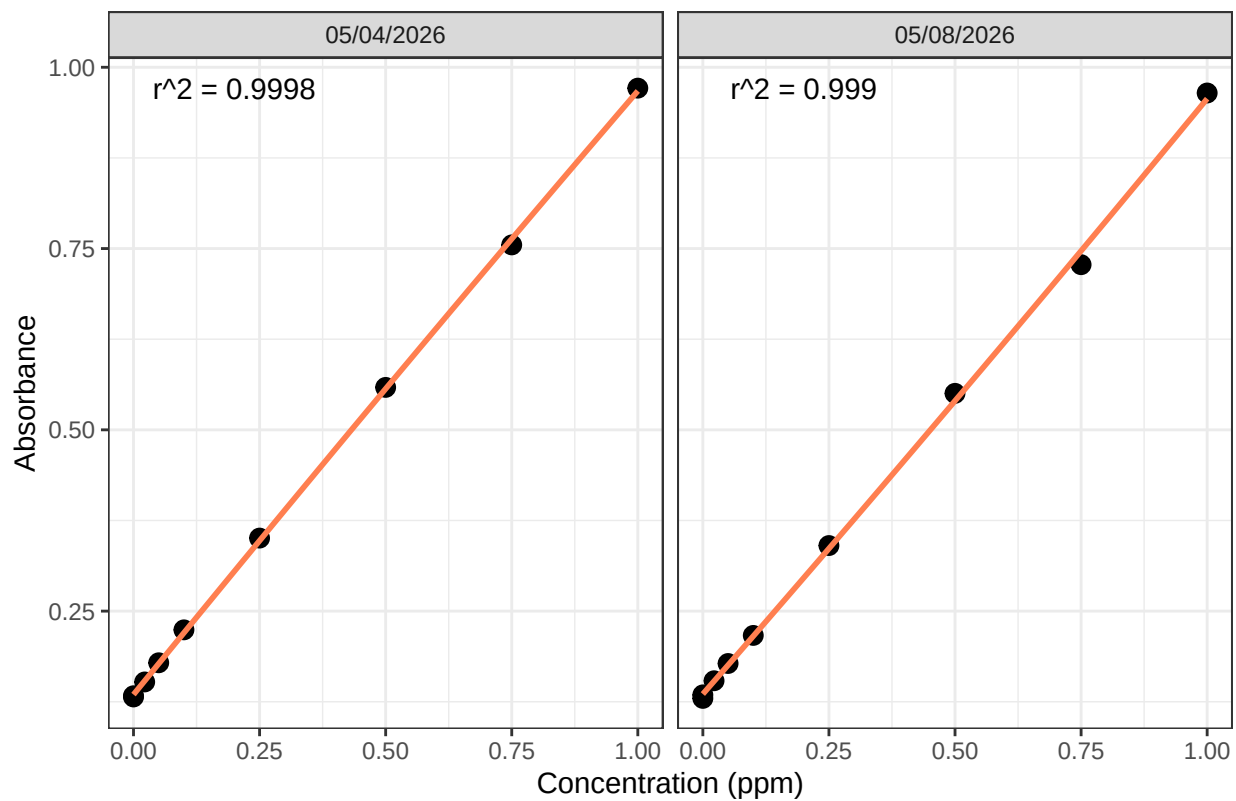


```
## `geom_smooth()` using formula = 'y ~ x'
```

PO4 Standard Curve

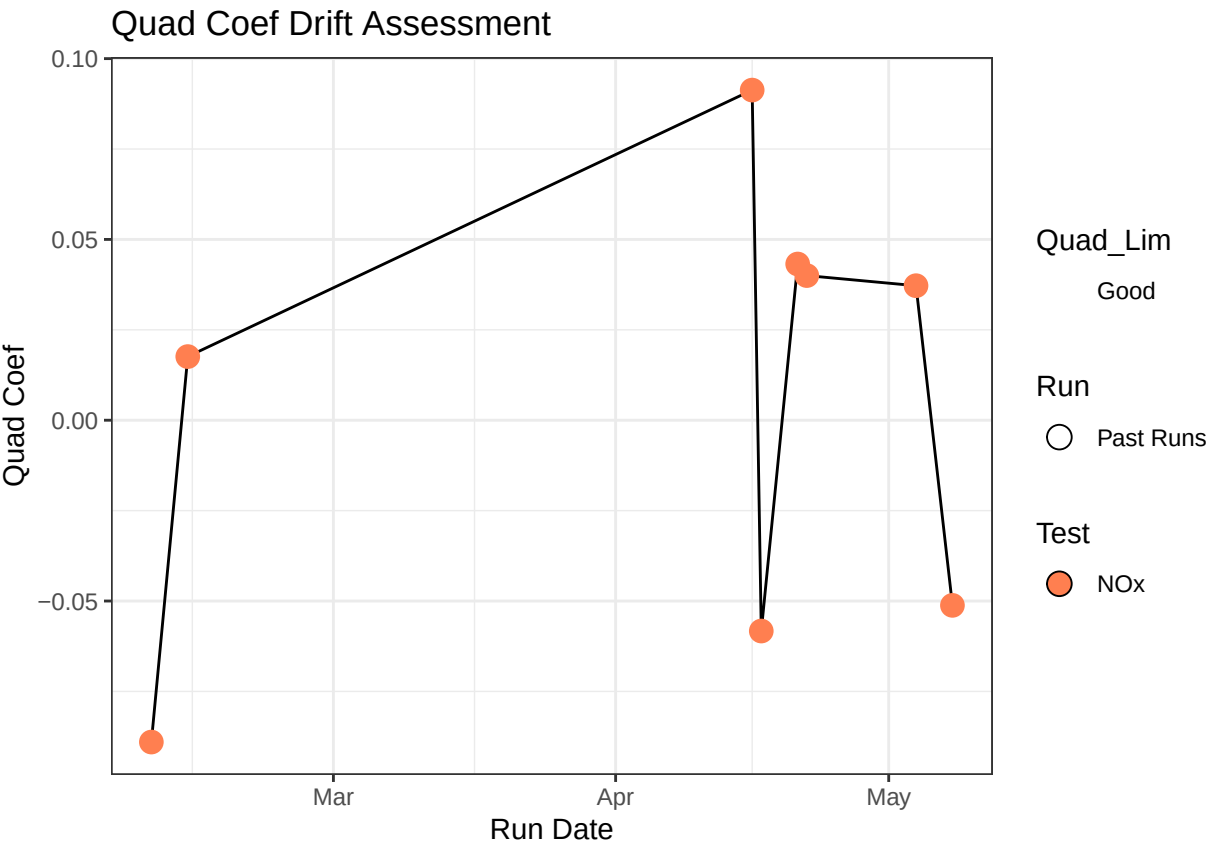
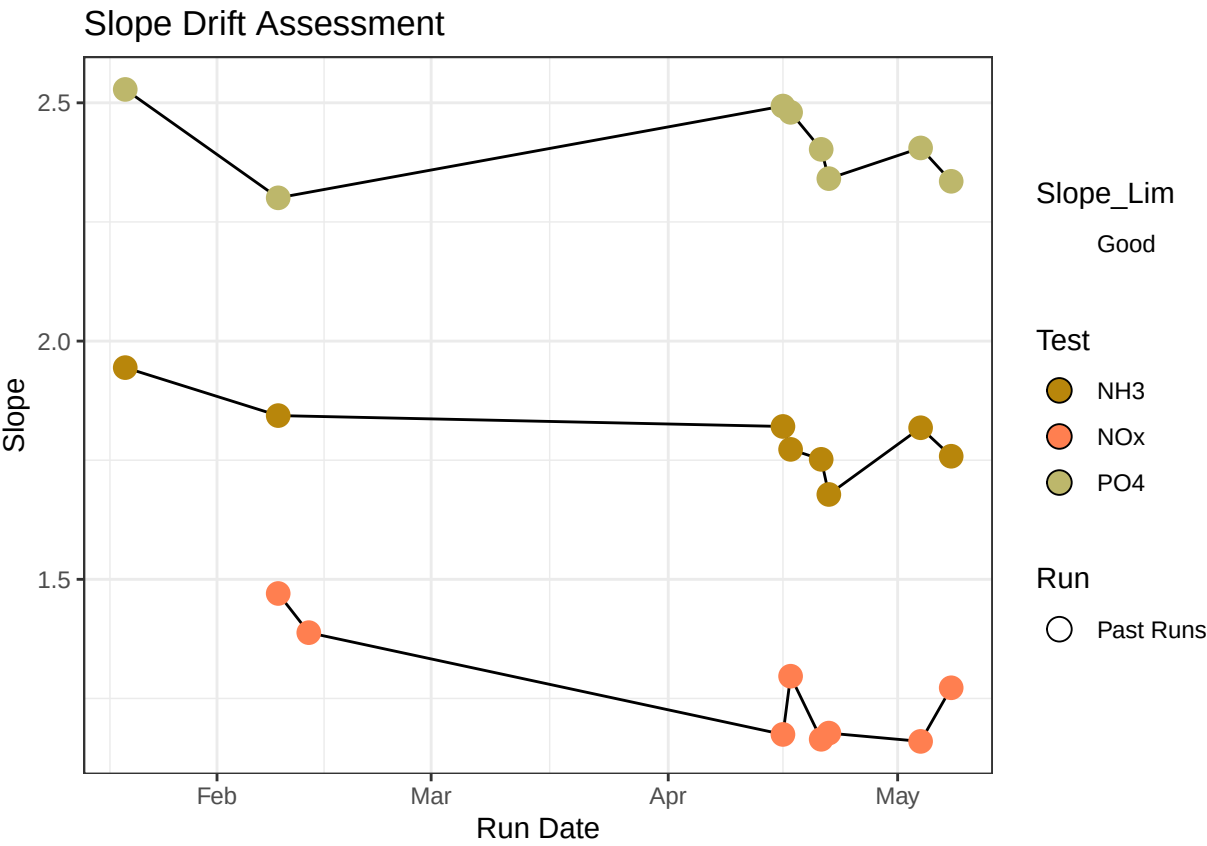


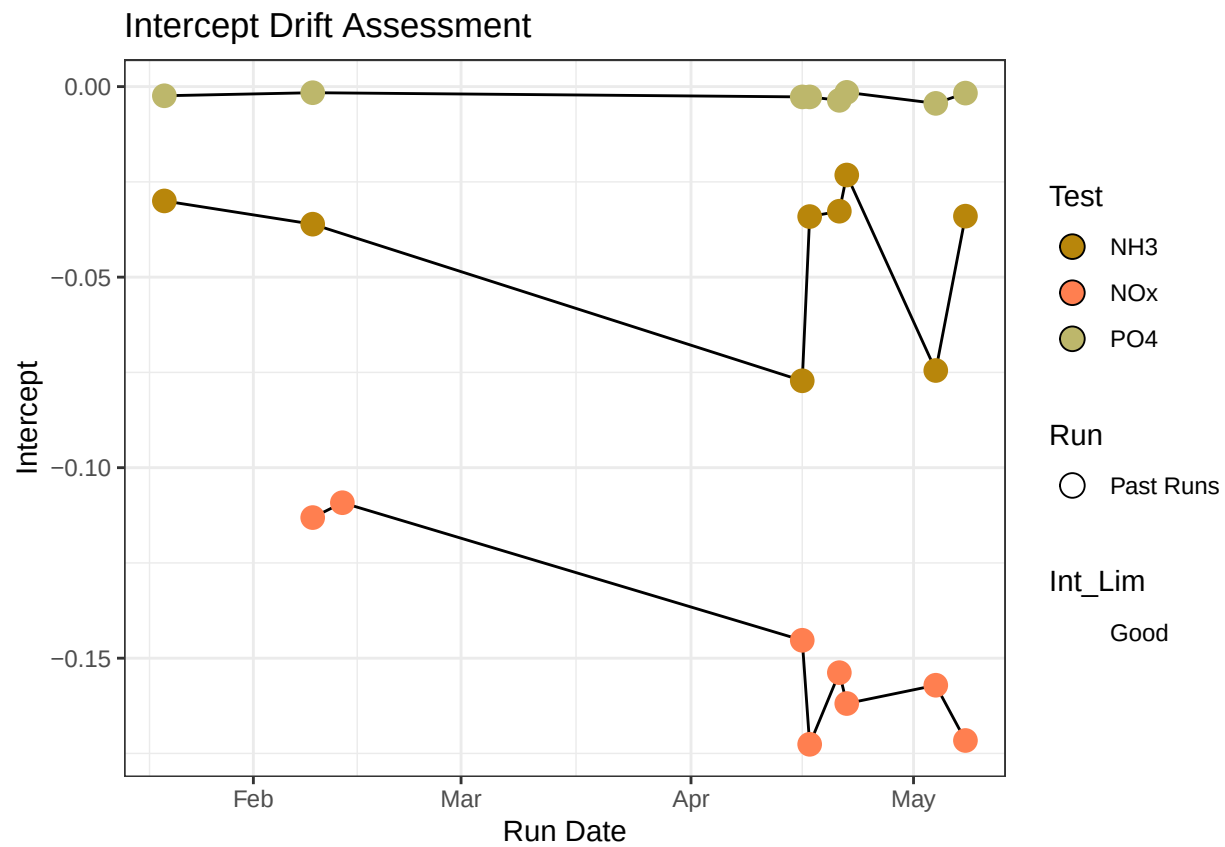
NOx Standard Curve



Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	05/04/2026	R2_Pass	0.9995982	1.818107
Ammonia 2	05/08/2026	R2_Pass	0.9999344	1.758303
o-PHOS 0.3	05/04/2026	R2_Pass	0.9993257	2.405344
o-PHOS 0.3	05/08/2026	R2_Pass	0.9993370	2.335316
Vanadium NOx	05/04/2026	R2_Pass	0.9997660	1.159949
Vanadium NOx	05/08/2026	R2_Pass	0.9989603	1.272420

0.3 Update Standard Log





0.4 Dilution Corrections - ensure the latest dilution is kept

```
## [1] "Reran Samples: TMP_SW_D5_20251101"
```

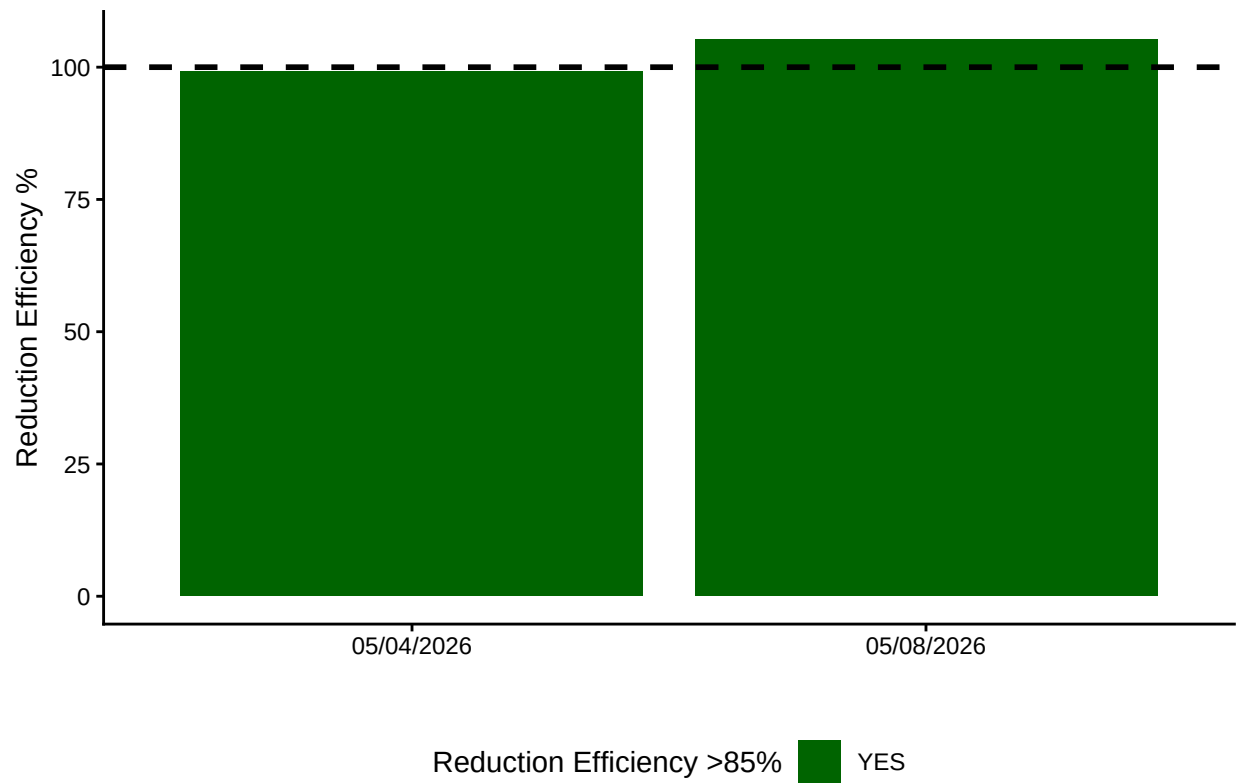
```
## [1] "Diluted Samples: TMP_SW_D5_20251101"
```

```
## [1] "No Naming Issues Detected"
```

0.5 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	05/04/2026	Performance Check Within 25% - PROCEED	1.2530565	1.034
Ammonia 2	05/08/2026	Performance Check Within 25% - PROCEED	1.2651620	1.034
o-PHOS 0.3	05/04/2026	Performance Check Within 25% - PROCEED	0.9222755	0.824
o-PHOS 0.3	05/08/2026	Performance Check Within 25% - PROCEED	0.9105505	0.824
Vanadium NOx	05/04/2026	Performance Check Within 25% - PROCEED	1.5966770	1.510
Vanadium NOx	05/08/2026	Performance Check Within 25% - PROCEED	1.6844615	1.510

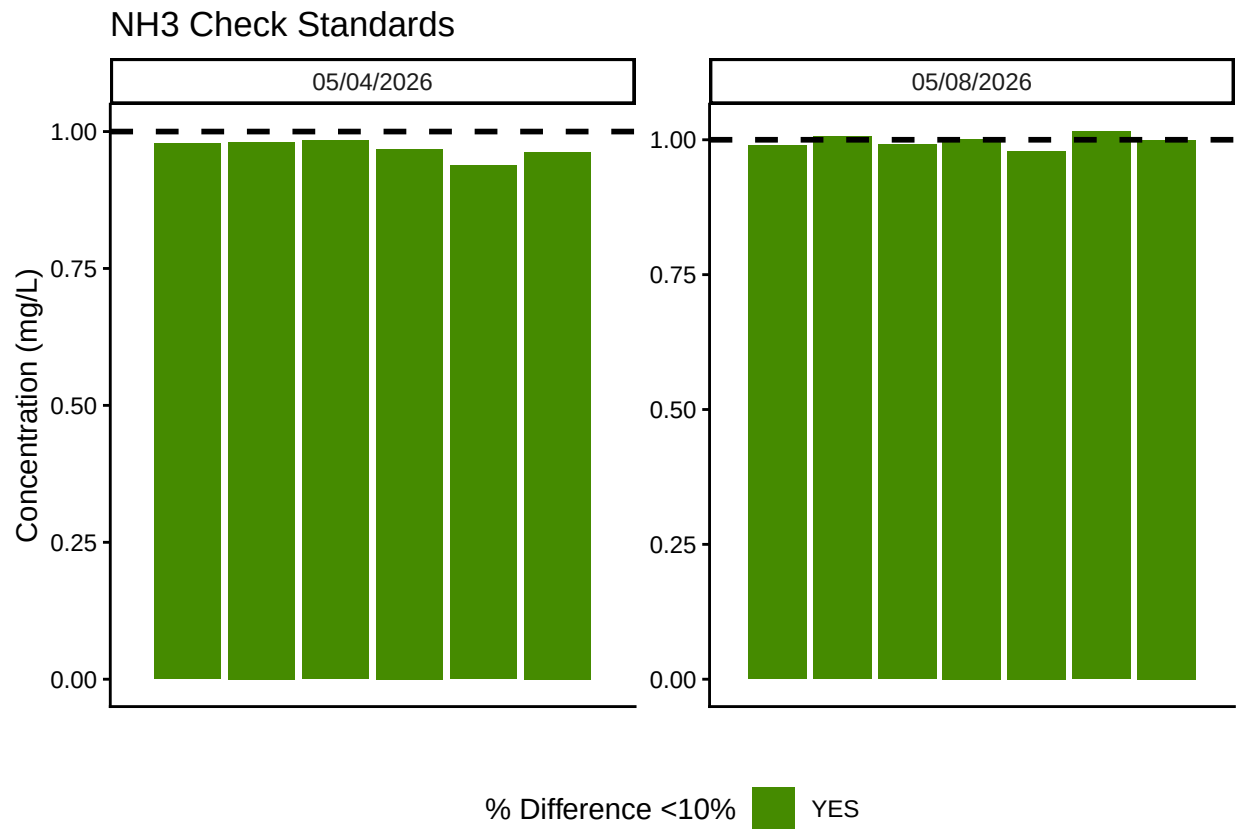
0.6 Check NOx Reduction Efficiency



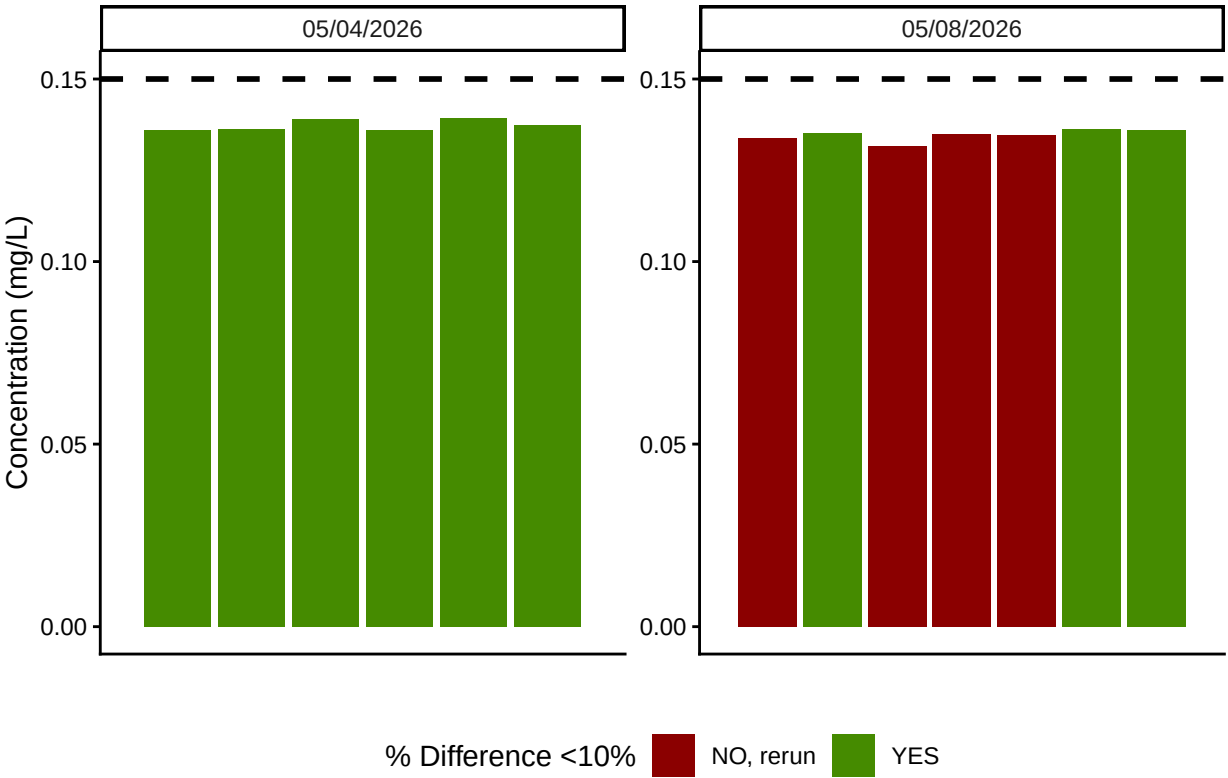
Test	Run_Date	Red_Eff_Flag
	05/04/2026	Mean NOx Reduction Efficiency >85% - PROCEED
	05/08/2026	Mean NOx Reduction Efficiency >85% - PROCEED

0.7 Analyze the Check Standards

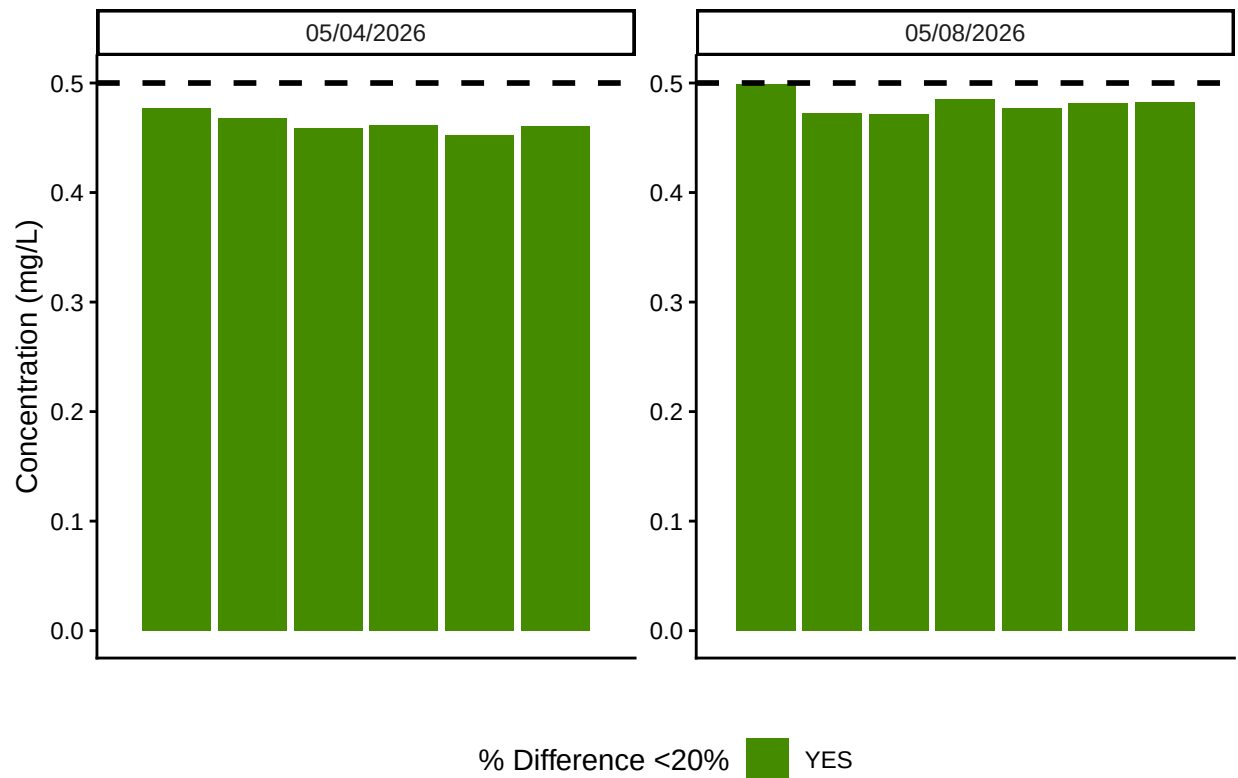
Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	05/04/2026	RSV WITHIN RANGE - PROCEED	0.0207134	0.25
Ammonia 2	05/08/2026	RSV WITHIN RANGE - PROCEED	0.0207134	0.25
o-PHOS 0.3	05/04/2026	RSV WITHIN RANGE - PROCEED	0.0150140	0.25
o-PHOS 0.3	05/08/2026	RSV WITHIN RANGE - PROCEED	0.0150140	0.25
Vanadium NOx	05/04/2026	RSV WITHIN RANGE - PROCEED	0.0269662	0.25
Vanadium NOx	05/08/2026	RSV WITHIN RANGE - PROCEED	0.0269662	0.25



PO4 Check Standards



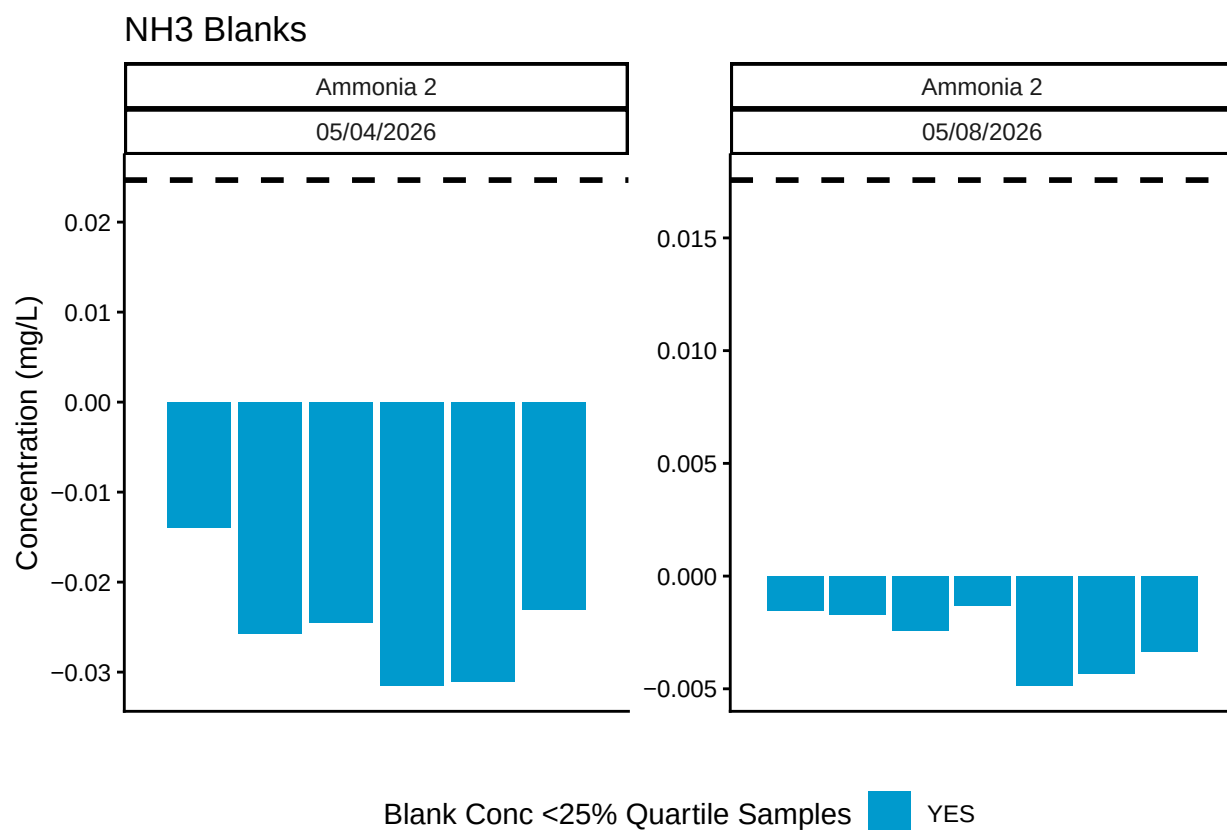
NOx Check Standards



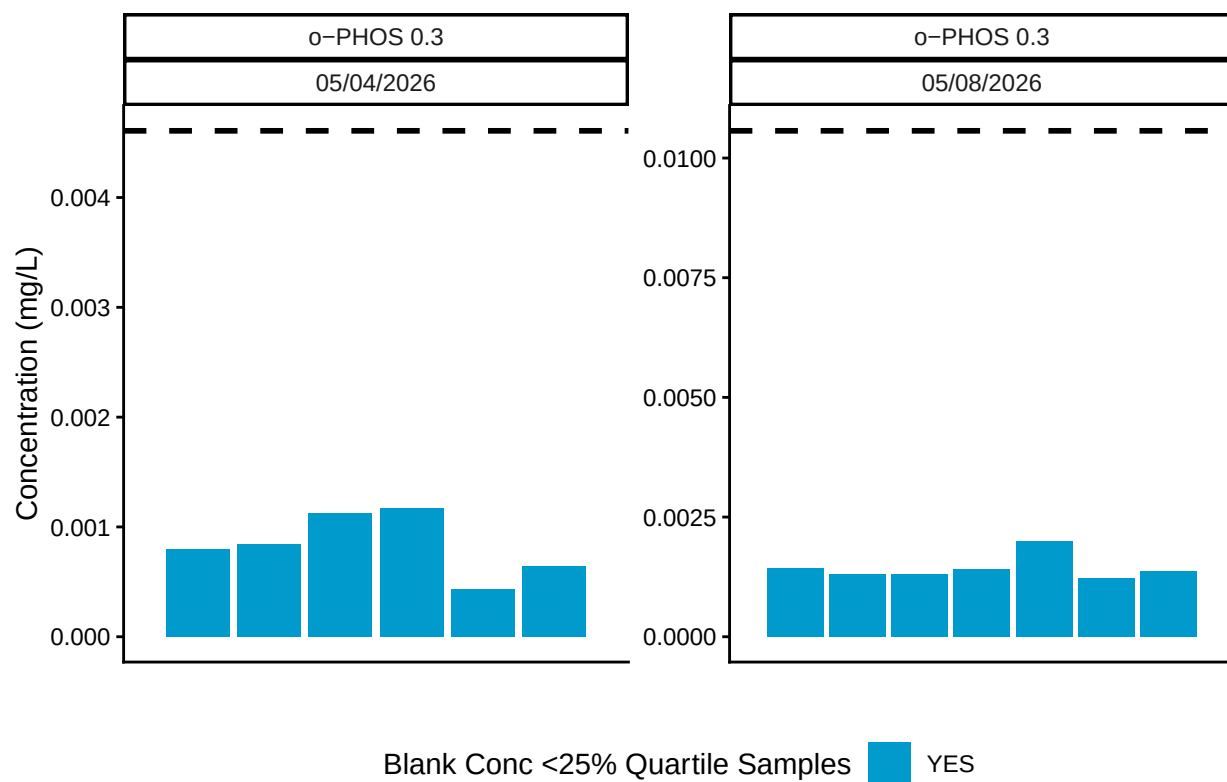
Test	Run_Date	CHK_Flag
Ammonia 2	05/04/2026	>60% of Checks Pass - PROCEED
Ammonia 2	05/08/2026	>60% of Checks Pass - PROCEED
o-PHOS 0.3	05/04/2026	>60% of Checks Pass - PROCEED
o-PHOS 0.3	05/08/2026	<60% of Checks Pass - REASSESs
Vanadium NOx	05/04/2026	>60% of Checks Pass - PROCEED
Vanadium NOx	05/08/2026	>60% of Checks Pass - PROCEED

0.8 Analyze Blanks

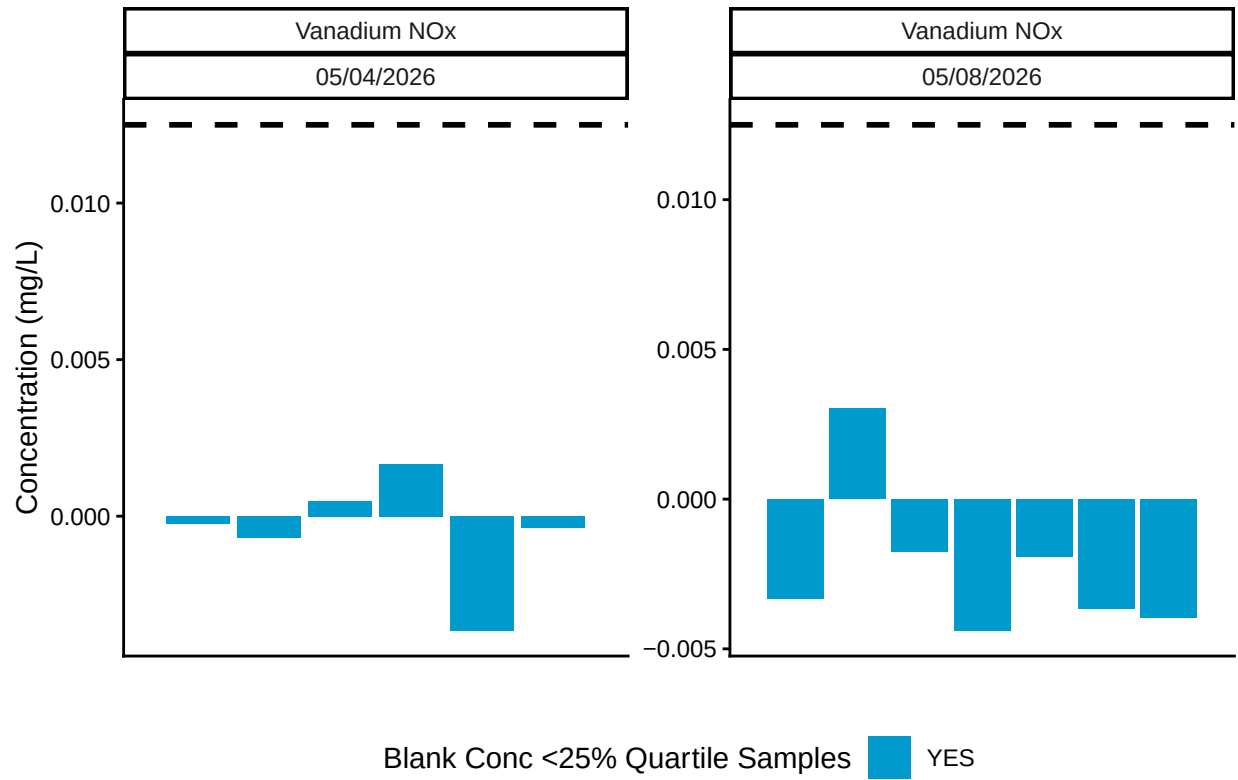
Test	Run_Date	BLK_Pct_Flag	Mean_Blkc_Conc	Quantile_25
Ammonia 2	05/04/2026	>60% of Blanks Pass - PROCEED	-0.0249550	0.0246705
Ammonia 2	05/08/2026	>60% of Blanks Pass - PROCEED	-0.0027873	0.0175657
o-PHOS 0.3	05/04/2026	>60% of Blanks Pass - PROCEED	0.0008352	0.0046075
o-PHOS 0.3	05/08/2026	>60% of Blanks Pass - PROCEED	0.0014364	0.0105687
Vanadium NOx	05/04/2026	>60% of Blanks Pass - PROCEED	-0.0004657	0.0125000
Vanadium NOx	05/08/2026	>60% of Blanks Pass - PROCEED	-0.0022839	0.0125000



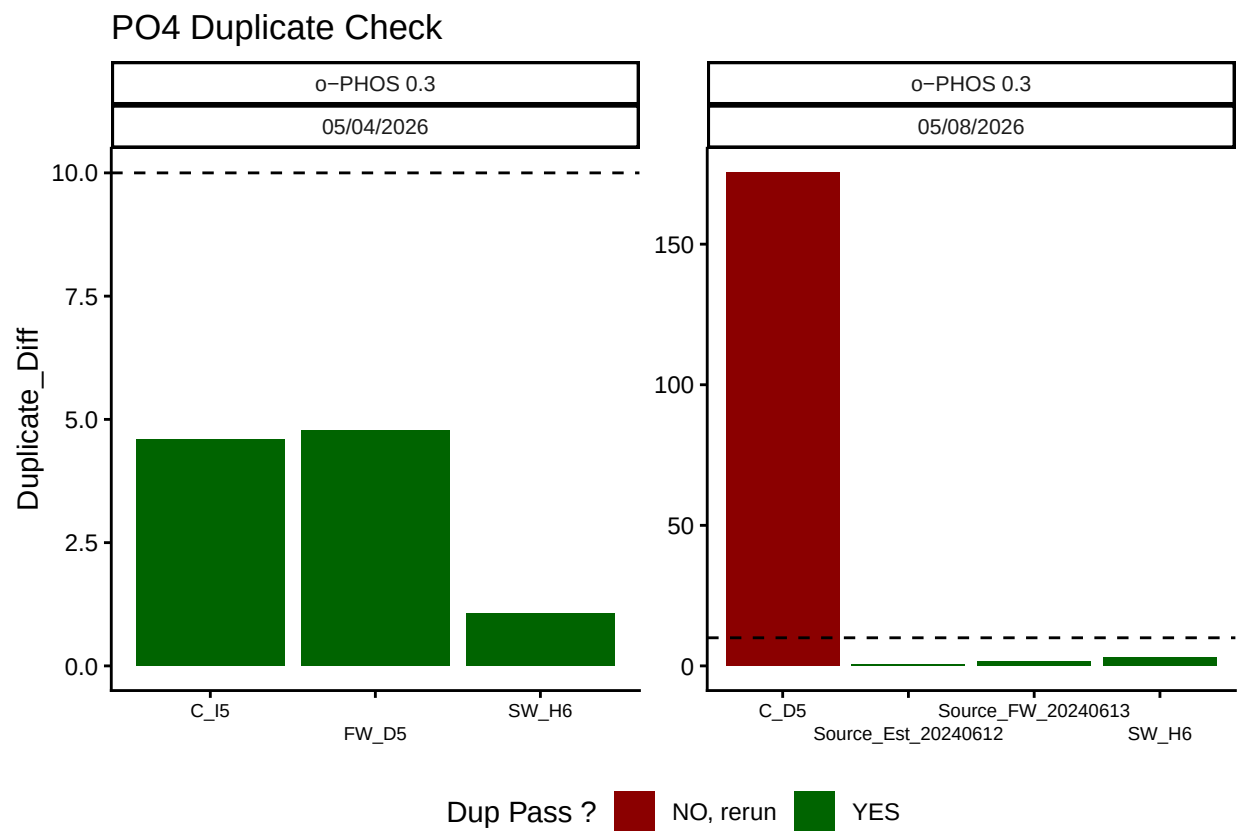
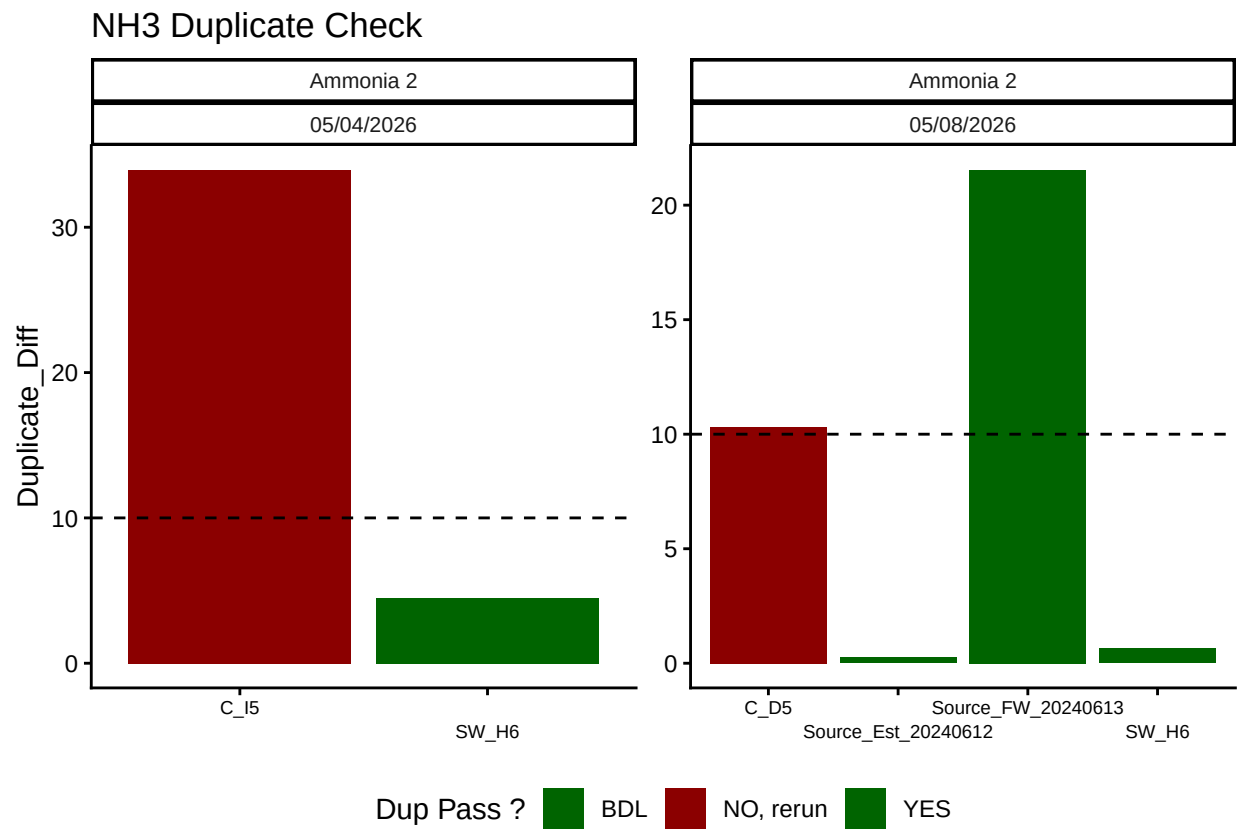
PO4 Blanks



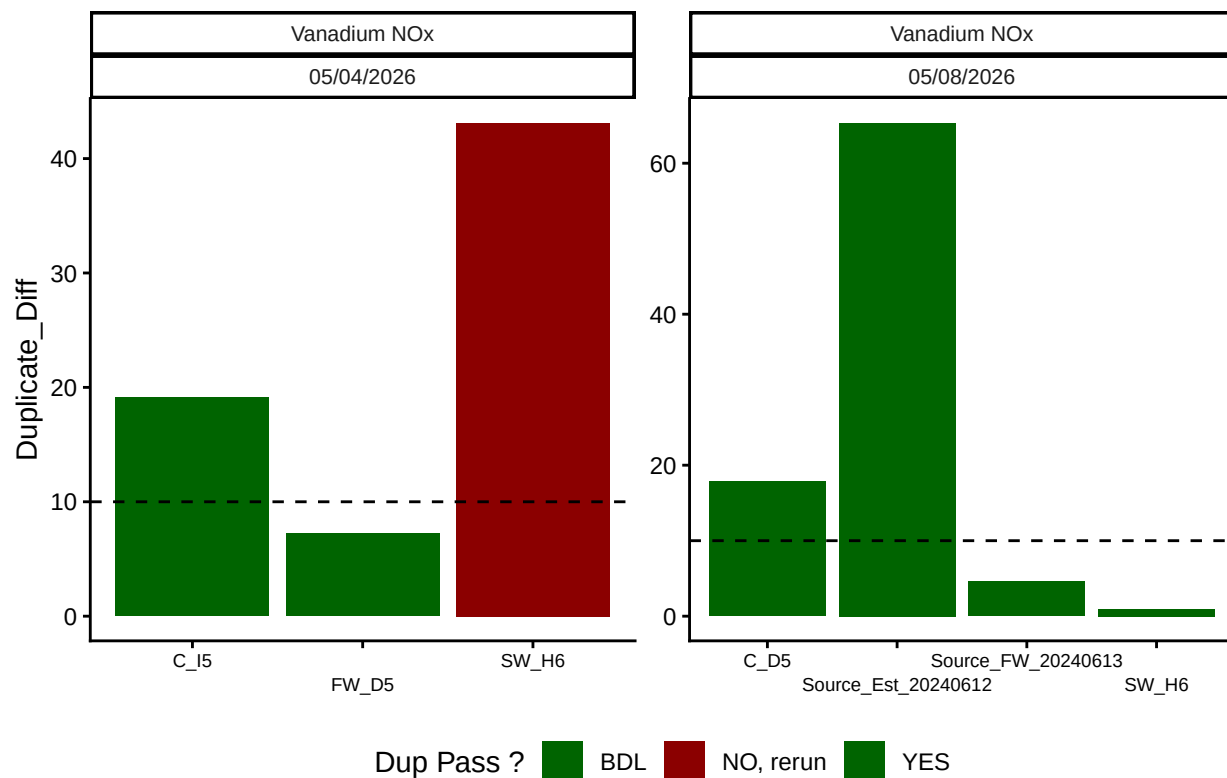
NOx Blanks



0.9 Analyze Duplicates

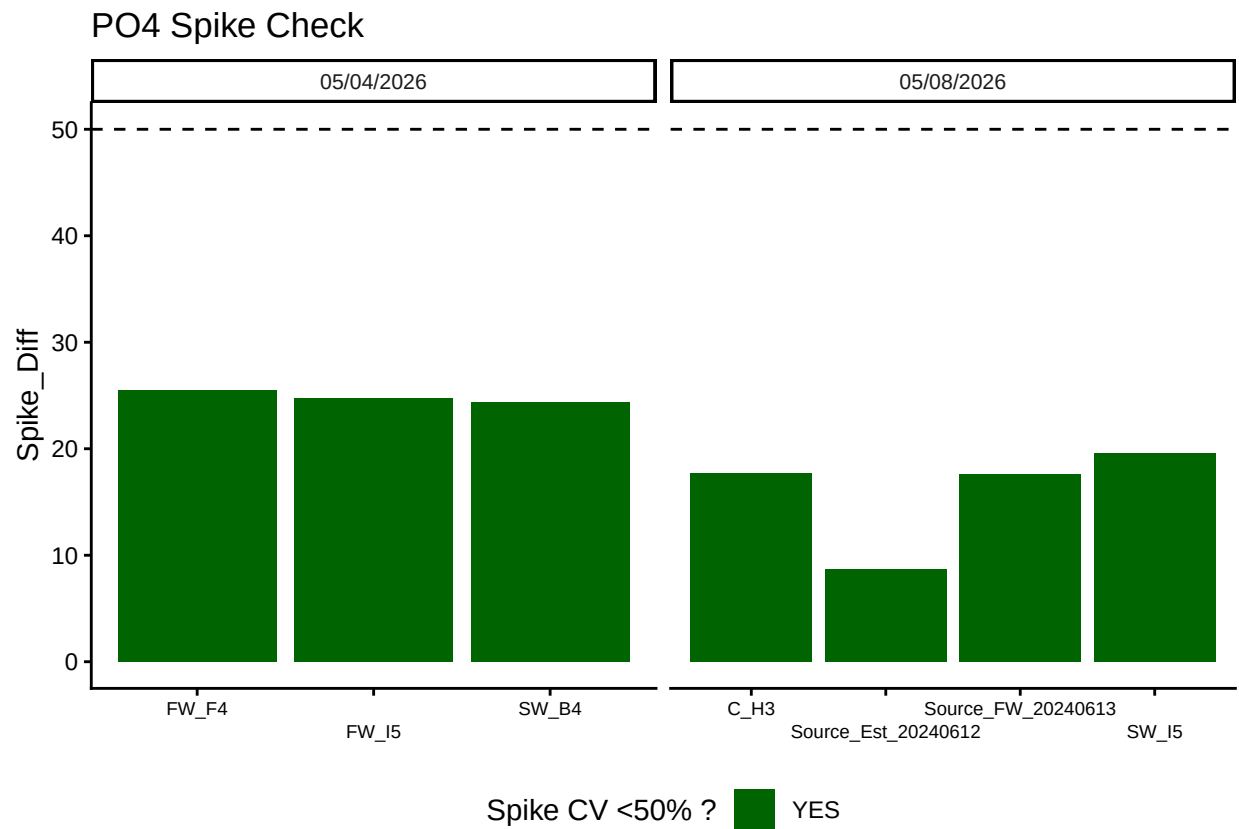
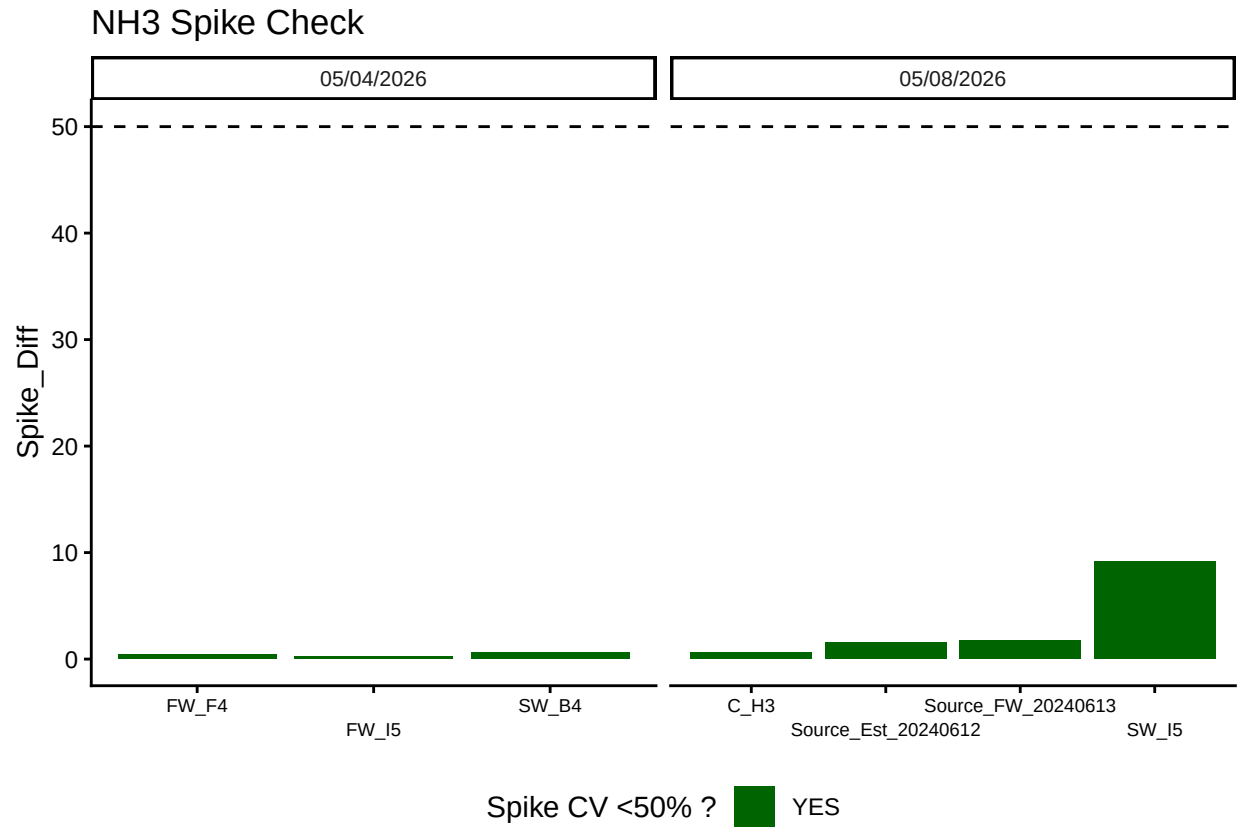


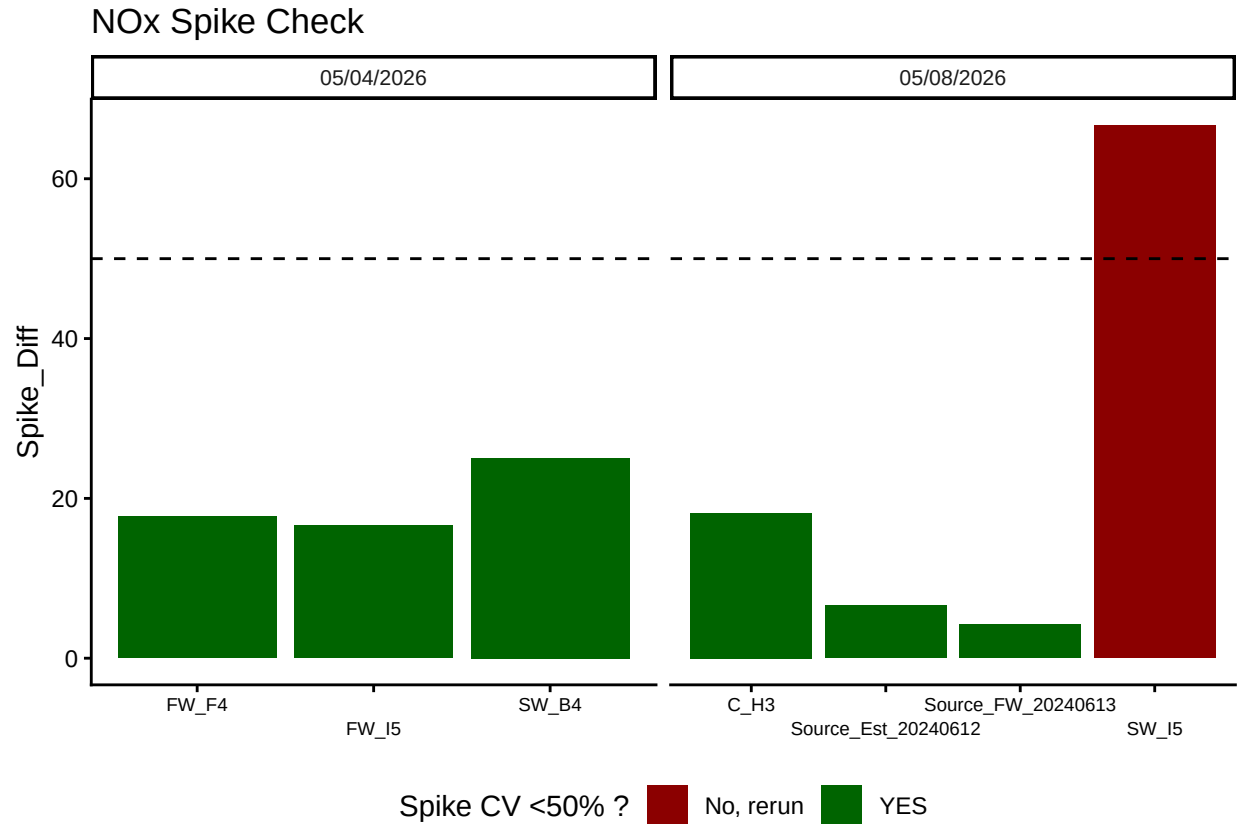
NOx Duplicate Check



Test	Run_Date	Dup_Flags
Ammonia 2	05/04/2026	<60% of Dups Pass - REASSESS
Ammonia 2	05/08/2026	<60% of Dups Pass - REASSESS
o-PHOS 0.3	05/04/2026	>60% of Dups Pass - PROCEED
o-PHOS 0.3	05/08/2026	>60% of Dups Pass - PROCEED
Vanadium NOx	05/04/2026	<60% of Dups Pass - REASSESS
Vanadium NOx	05/08/2026	<60% of Dups Pass - REASSESS

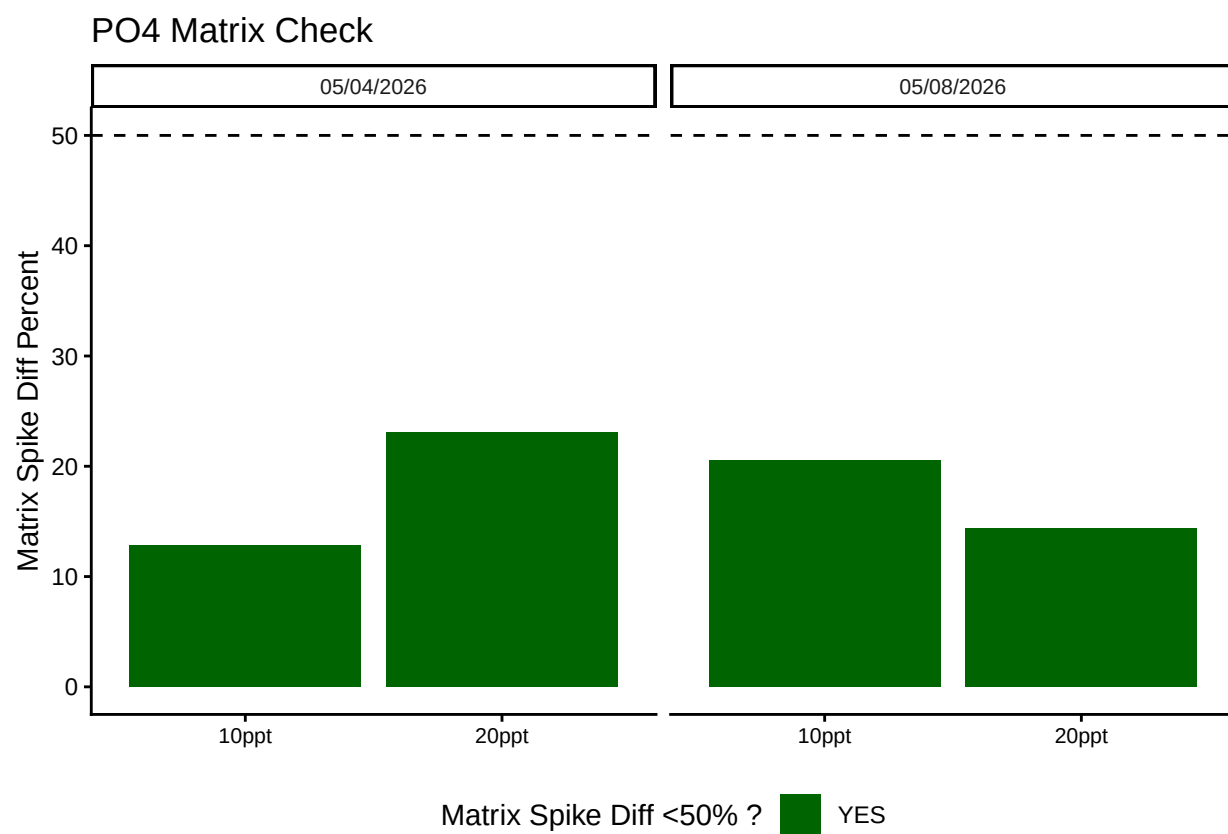
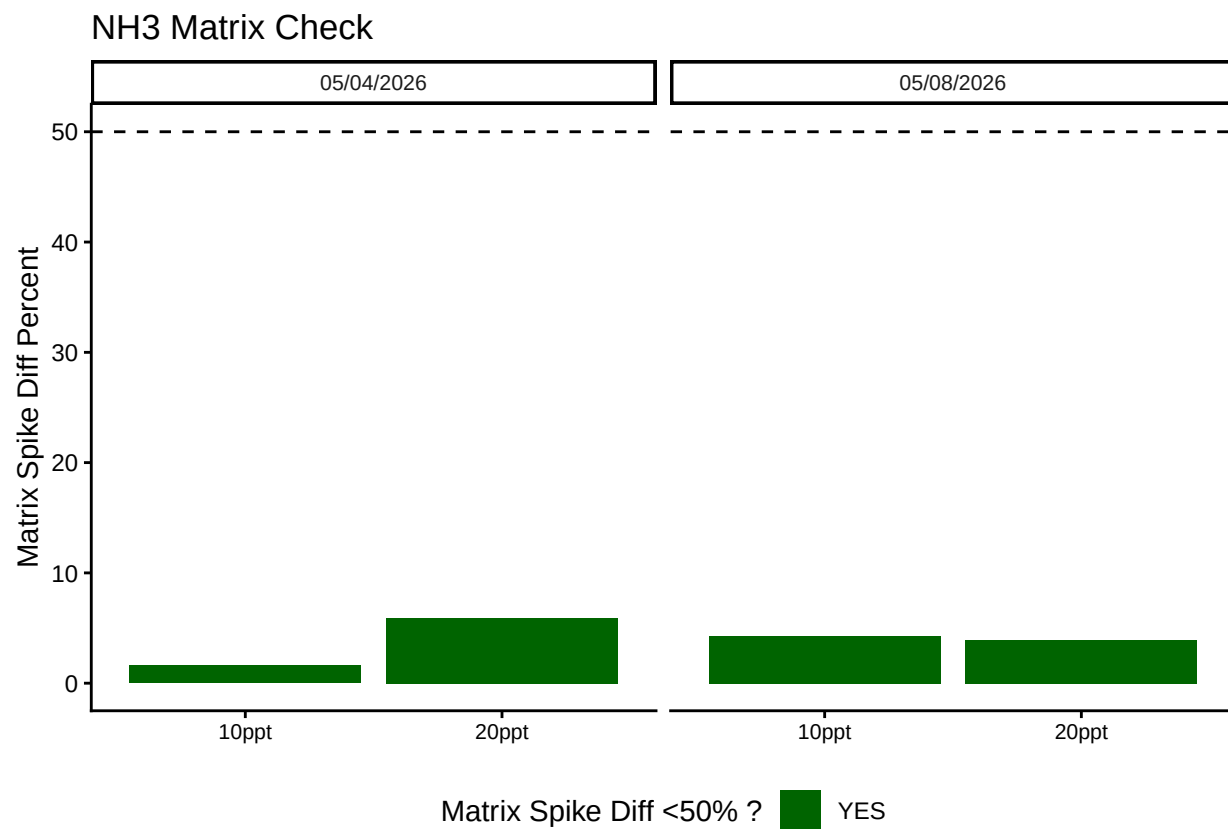
0.10 Spikes

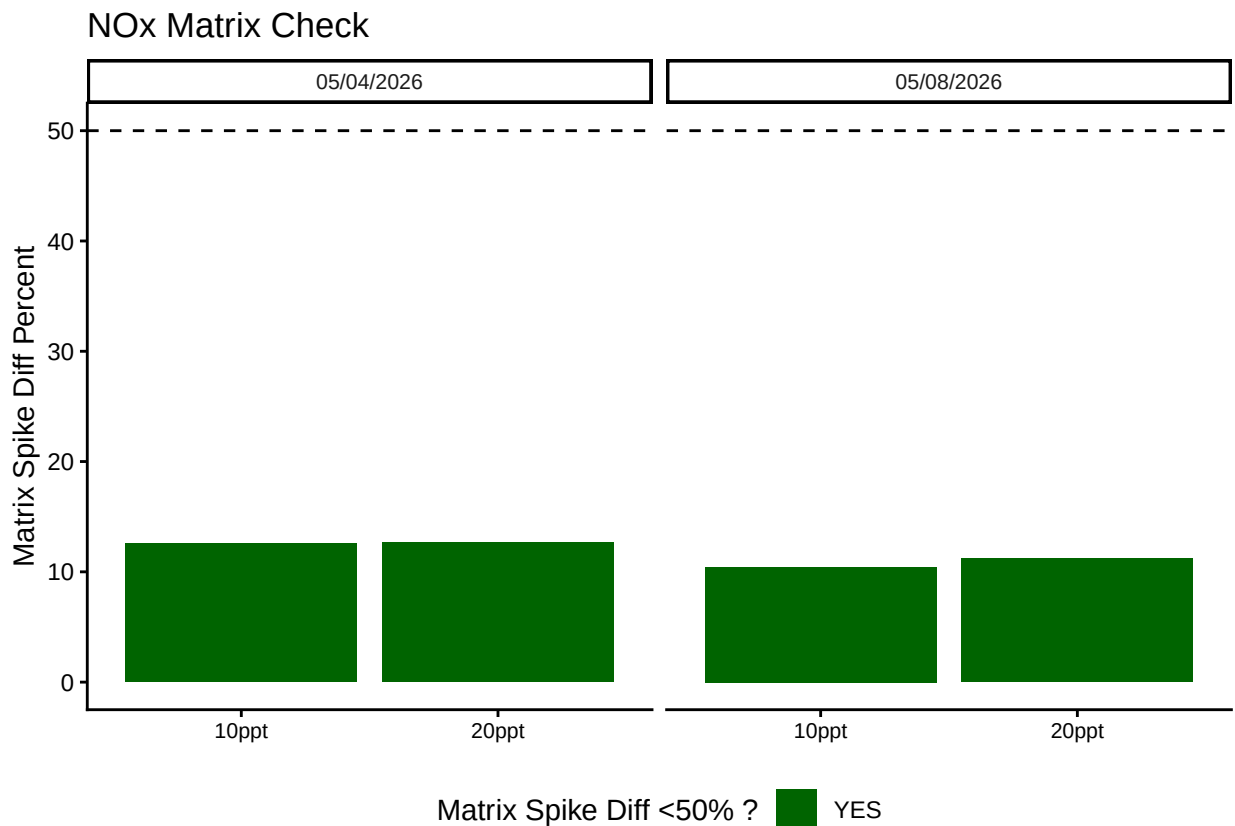




Test	Run_Date	Spike_Flags
Ammonia 2	05/04/2026	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	05/08/2026	>60% of Spikes have a Diff <50% - PROCEED
Vanadium NOx	05/04/2026	>60% of Spikes have a Diff <50% - PROCEED
Vanadium NOx	05/08/2026	>60% of Spikes have a Diff <50% - PROCEED
o-PHOS 0.3	05/04/2026	>60% of Spikes have a Diff <50% - PROCEED
o-PHOS 0.3	05/08/2026	>60% of Spikes have a Diff <50% - PROCEED

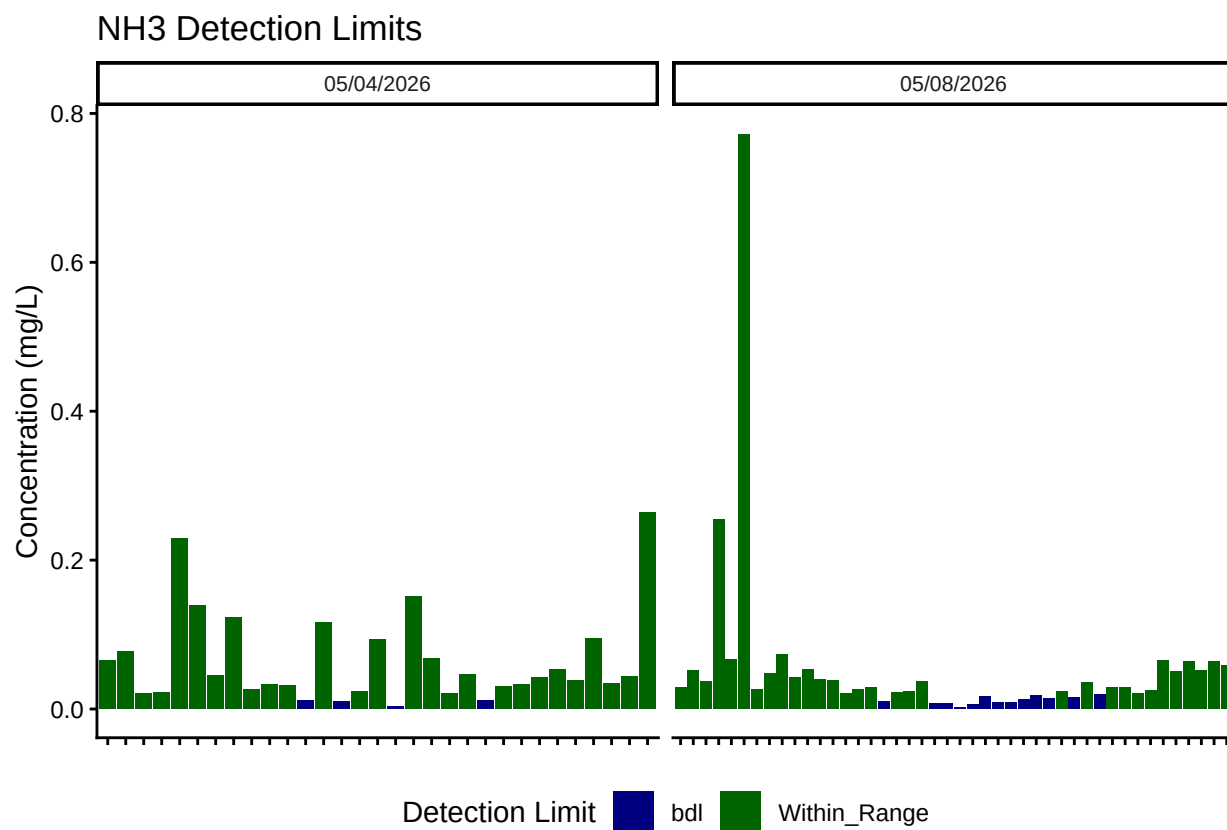
0.11 Matrix Effects

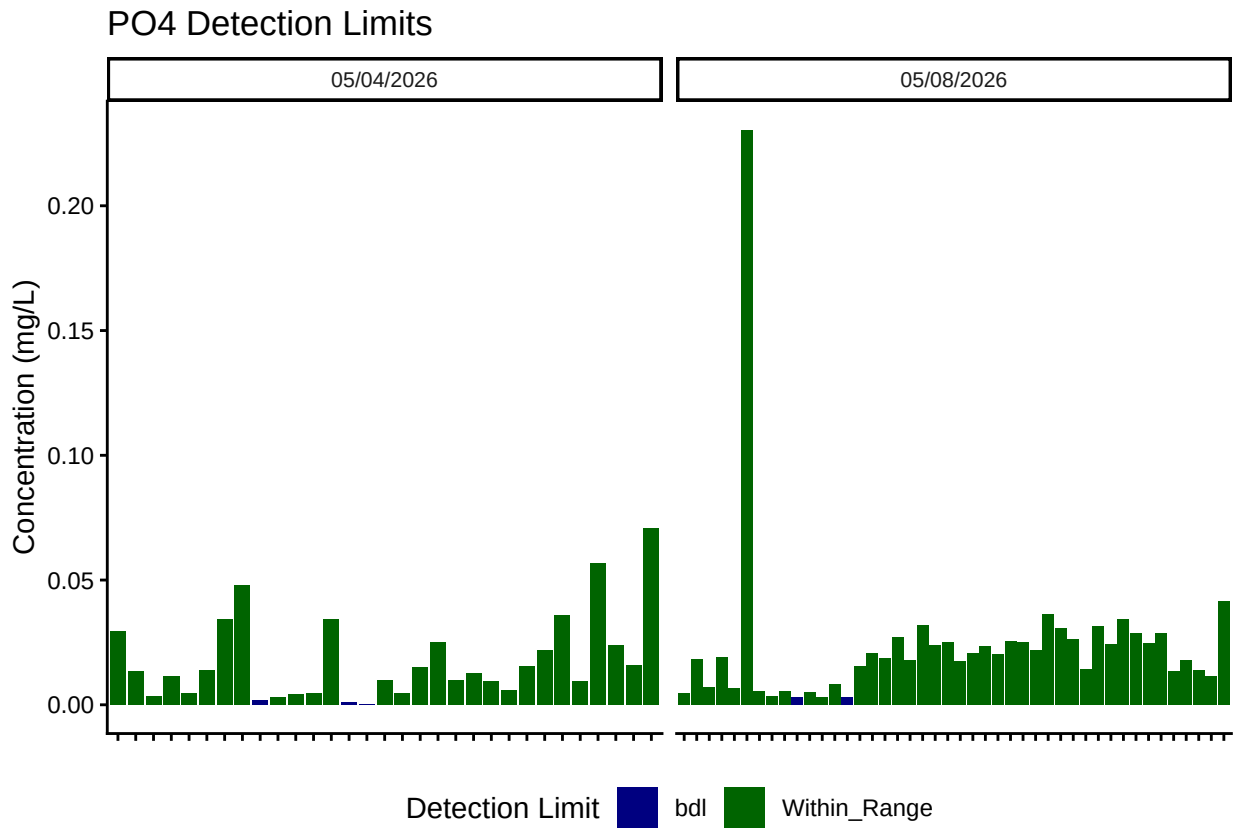


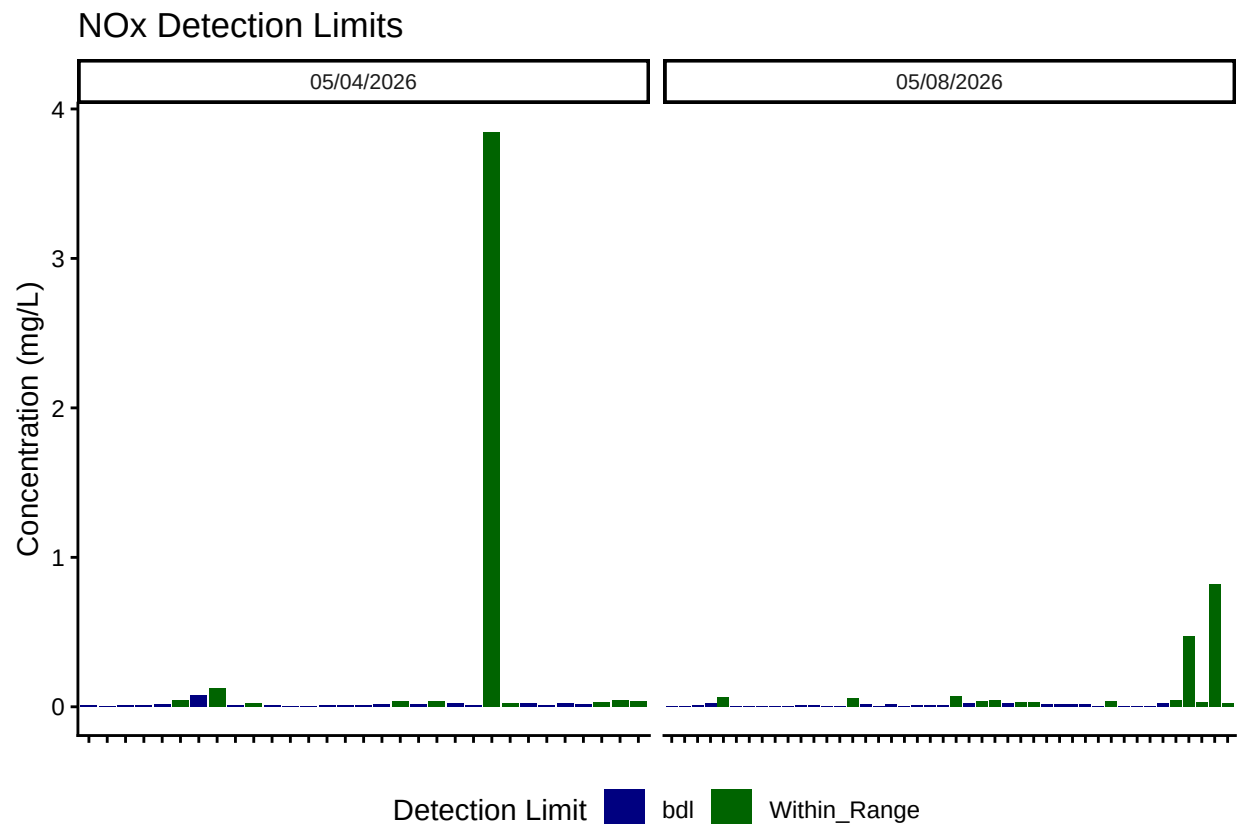


Test	Run_Date	Matrix_Flags
Ammonia 2	05/04/2026	Matrix Has Diff <50% - PROCEED
Ammonia 2	05/08/2026	Matrix Has Diff <50% - PROCEED
Vanadium NOx	05/04/2026	Matrix Has Diff <50% - PROCEED
Vanadium NOx	05/08/2026	Matrix Has Diff <50% - PROCEED
o-PHOS 0.3	05/04/2026	Matrix Has Diff <50% - PROCEED
o-PHOS 0.3	05/08/2026	Matrix Has Diff <50% - PROCEED

##Sample Flagging - Within range of standard curve







##Add QAQC Flags

0.12 Check to see if samples run match metadata & merge info

Some sample IDs are missing from metadata.

```
## [1] "TMP_Source_Est_20240611_LT" "TMP_Source_Est_20240611_HT"
## [3] "TMP_Source_FW_20240611_MN" "TMP_Source_FW_20240611_MD"
## [5] "TMP_Source_FW_20240611_AN" "TMP_Source_SW_20240611_MN"
## [7] "TMP_Source_SW_20240611_MD" "TMP_Source_SW_20240611_AN"
## [9] "TMP_Source_Est_20240612_LT" "TMP_Source_Est_20240612_HT"
## [11] "TMP_Source_FW_20240612_MN" "TMP_Source_FW_20240612_MD"
## [13] "TMP_Source_FW_20240612_AN" "TMP_Source_SW_20240612_MN"
## [15] "TMP_Source_SW_20240612_MD" "TMP_Source_SW_20240612_AN"
## [17] "TMP_Source_Est_20240613_LT" "TMP_Source_Est_20240613_HT"
## [19] "TMP_Source_FW_20240613_MN" "TMP_Source_FW_20240613_MD"
## [21] "TMP_Source_FW_20240613_AN" "TMP_Source_SW_20240613_MN"
## [23] "TMP_Source_SW_20240613_MD" "TMP_Source_SW_20240613_AN"
## [25] "TMP_FW_D5_20260309" "TMP_FW_F4_20260309"
## [27] "TMP_FW_F6_20260309" "TMP_FW_H3_20260309"
## [29] "TMP_C_H6_20260602" "TMP_SW_F6_20260309"
## [31] "TMP_SW_H6_20260309" "TMP_SW_I5_20260309"
## [33] "TMP_FW_B4_20260309" "TMP_FW_C6_20260309"
## [35] "TMP_C_H6_20260309" "TMP_C_I5_20260309"
## [37] "TMP_SW_B4_20260309" "TMP_SW_C6_20260309"
```

```
## [39] "TMP_SW_D5_20260309"      "TMP_FW_D5_20260410"
## [41] "TMP_C_B4_20260309"      "TMP_C_C3_20260309"
## [43] "TMP_C_D5_20260309"      "TMP_C_H3_20260309"
```

```
## [1] "Sample_Name" "Run_Number" "Conc"      "Absorbance" "Dilution"
## [6] "Unit"        "Test"       "Run_Time"  "Run_Date"   "Keep"
## [11] "Pair_ID"     "Conc_uM"    "Conc_flag"
```

```
##Format Data
```

0.13 Visualize Data by Plot

```
## Site_Code Plot Grid_Square Date
## 1      TMP      C      B4 20260309
## 2      TMP      C      C3 20260309
## 3      TMP      C      D5 20260309
## 4      TMP      C      F6 20250711
## 5      TMP      C      H3 20250711
## 6      TMP      C      H3 20260309
```

```
## Site_Code Plot Grid_Square Date Sample_ID Site Zone Grid
## 1      TMP      C      B4 20260309 TMP_C_B4_20260309 Tempest <NA> <NA>
## 2      TMP      C      C3 20260309 TMP_C_C3_20260309 Tempest <NA> <NA>
## 3      TMP      C      D5 20260309 TMP_C_D5_20260309 Tempest <NA> <NA>
## 4      TMP      C      F6 20250711 TMP_C_F6_20250711 Tempest      C      F6
## 5      TMP      C      H3 20250711 TMP_C_H3_20250711 Tempest      C      H3
## 6      TMP      C      H3 20260309 TMP_C_H3_20260309 Tempest <NA> <NA>
```

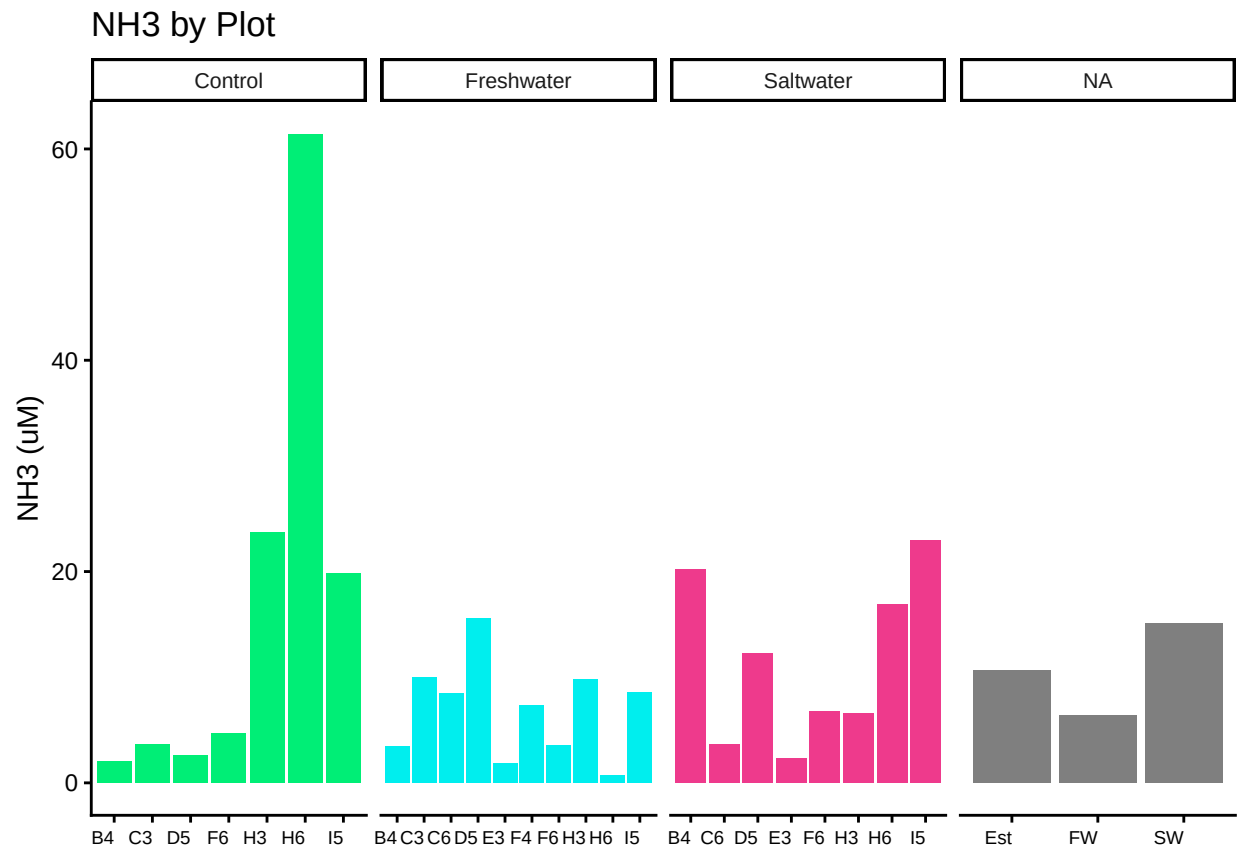
```
## Depth_cm Event_Time Time_of_day Source Volume_mL Collection_Date
## 1      <NA>      <NA>      <NA>      <NA>      NA      <NA>
## 2      <NA>      <NA>      <NA>      <NA>      NA      <NA>
## 3      <NA>      <NA>      <NA>      <NA>      NA      <NA>
## 4      15      <NA>      <NA>      PW      7.5      20250711
## 5      15      <NA>      <NA>      PW      14.0     20250711
## 6      <NA>      <NA>      <NA>      <NA>      NA      <NA>
```

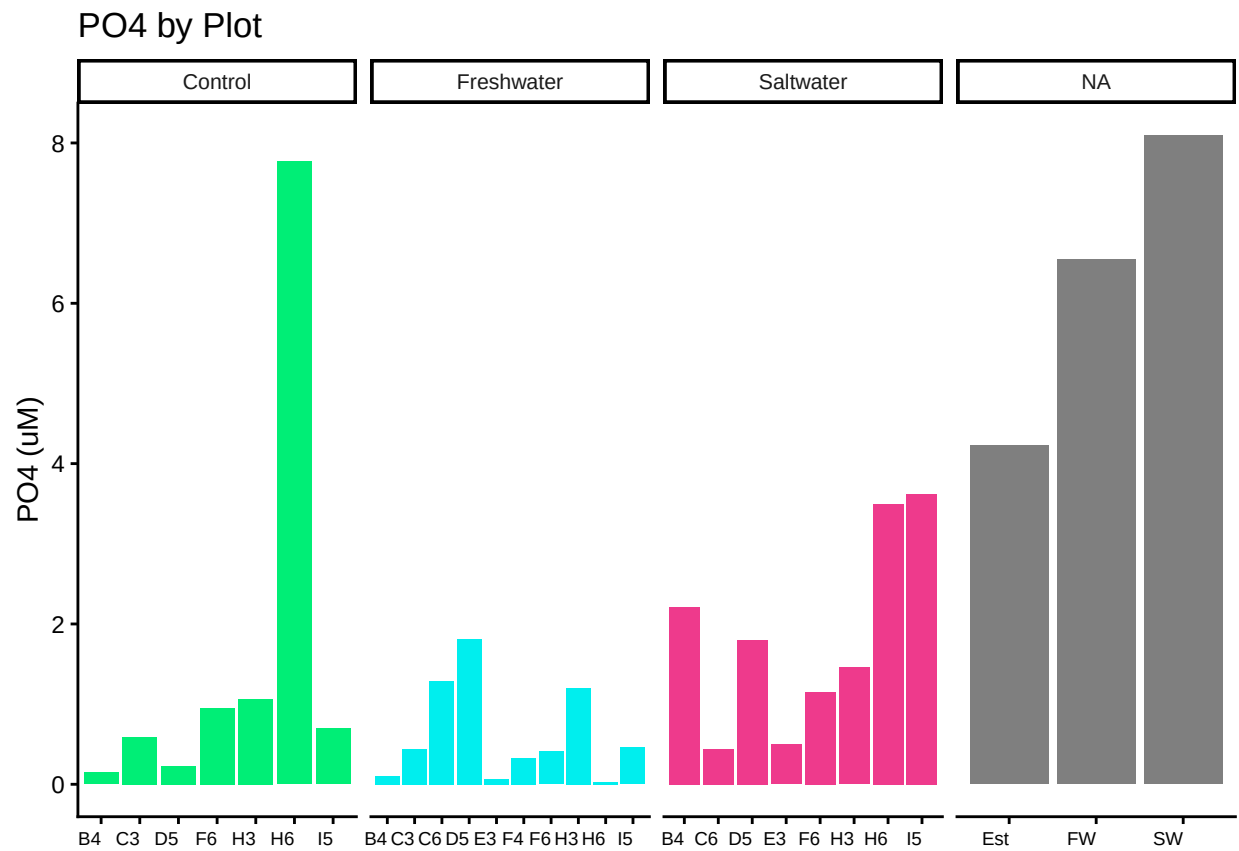
```
## Evacuation_Date NOx_Conc_mgL NOx_Conc_uM NOx_Conc_Flag NOx_QAQC_Flag
## 1      NA      0.002901 0.2071152      bdl
## 2      NA      0.001754 0.1252258      bdl
## 3      NA      0.012405 0.8856476      bdl
## 4      20250708 0.007791 0.5562338      bdl
## 5      20250708 0.002212 0.1579244      bdl
## 6      NA      0.024258 1.7318855      bdl
```

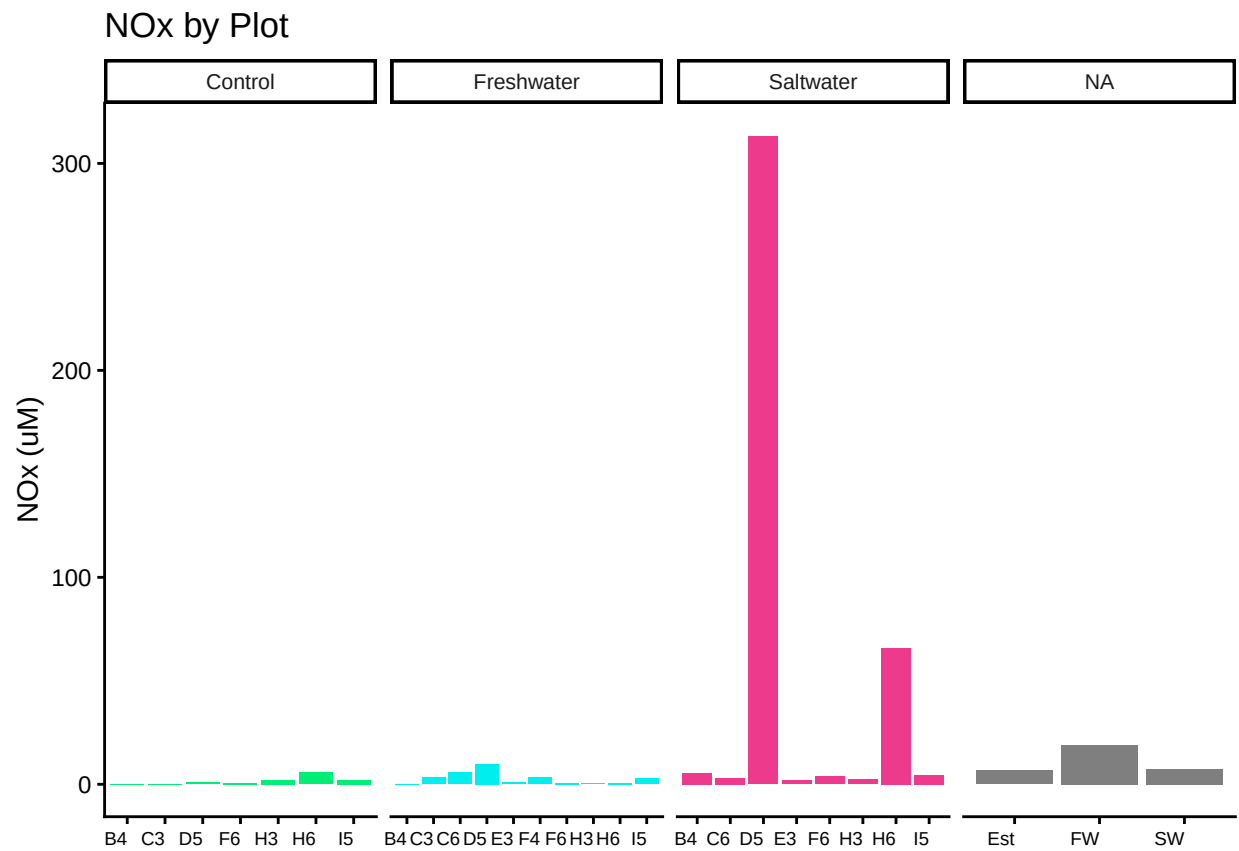
```
## NH3_Conc_mgL NH3_Conc_uM NH3_Conc_Flag NH3_QAQC_Flag PO4_Conc_mgL PO4_Conc_uM
## 1      0.028899 2.063227 Within_Range      0.004516 0.1458008
## 2      0.051426 3.671529 Within_Range      0.018202 0.5876587
## 3      0.037136 2.651303 Within_Range      0.007024 0.2267726
## 4      0.065259 4.659127 Within_Range      0.029424 0.9499653
## 5      0.077177 5.510006 Within_Range      0.013632 0.4401144
## 6      0.255198 18.219709 Within_Range      0.019172 0.6189755
```

```
## PO4_Conc_Flag PO4_QAQC_Flag Analysis_rundate Run_notes Field_notes
## 1 Within_Range      05/08/2026      <NA>
## 2 Within_Range      05/08/2026      <NA>
## 3 Within_Range      05/08/2026      <NA>
```

## 4	Within_Range	05/04/2026	<NA>
## 5	Within_Range	05/04/2026	<NA>
## 6	Within_Range	05/08/2026	<NA>







##Write Out Data

#end